

Secure AWS Architecture — Mid-Level Cloud Engineer Design Document

Context

This document describes the design and operation of a private AWS environment built to support internal workloads with a focus on security, reliability, and cost awareness.

Requirements

- No public exposure
- Identity-based access
- Centralized logging
- Low operational overhead
- Easy teardown

Architecture

Private VPC with segmented subnets, controlled egress, IAM-based access, and centralized monitoring.

Design Decisions

Chose private subnets to reduce attack surface; IAM + SSM over SSH to eliminate key management; endpoints over public APIs for security.

Networking

CIDR segmented across AZs. No IGW on workloads. NAT used only for required outbound access.

Identity & Access

IAM roles attached to instances. Session Manager for access. Least privilege policies reviewed and scoped.

Security

Default deny inbound, restricted outbound, IMDSv2 enforced, logging enabled for all actions.

Observability

CloudTrail across regions. CloudWatch logs and metrics reviewed for errors and anomalies.

Operations

Provision → Validate → Monitor → Decommission. Infrastructure treated as reproducible and disposable.

Cost

NAT and endpoints incur cost; environment deleted when idle.

Risks & Mitigations

Misconfiguration risk mitigated with documentation and review. Outage risk mitigated via monitoring and reproducibility.

Results

Environment met security and operational goals with minimal complexity.

Conclusion

Demonstrates practical application of AWS security, networking, and operational best practices.